

Day 22 Quiz

Geometry Summer Credit | Scale Effects, Density, and Design Constraints

Name:	Date:	Score:
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Directions

Choose the best answer. Show work for short-response items.

1. If the scale factor between two similar figures is 5, the area scale factor is:
A 5 B 10 C 25 D 125
2. If the scale factor between two similar solids is 2, the volume scale factor is:
A 2 B 4 C 6 D 8
3. Density is calculated by:
A volume / mass B mass / volume C area / volume D perimeter / area
4. A design constraint is best described as:
A a random measurement B a required limit or condition C only a formula D only a drawing
5. A model has scale factor $\frac{1}{4}$ of the original. A 40-foot actual length becomes:
A 4 ft B 10 ft C 16 ft D 160 ft
6. Two similar solids have scale factor 3. The smaller surface area is 50 square units. Find the larger surface area.
7. Two similar solids have scale factor 3. The smaller volume is 50 cubic units. Find the larger volume.

Day 22 Quiz

Continued

8.	A solid has mass 900 g and volume 300 cm^3 . Find density.
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9.	A tank is 9 ft by 5 ft by 4 ft. Does it meet a minimum capacity of 175 cubic feet? Explain.
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10.	Error analysis: A student uses k^2 to scale volume. Explain the error and give the correct scale factor for volume.
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